92 AM collinear spin layer groups

Néel directions, SOC magnetic layer groups and layer-resolved intrinsic nonlinear Hall constraints

Coverage. All 92 entries of Table S7, in the original order, expanded into **774 direction branches**: 15 groups have 3 branches, 30 have 7, 6 have 8, 36 have 11, and 5 have 15. Directions related by a parent symmetry are retained in the fixed source coordinate setting so that the displayed tensor indices stay meaningful. These 774 rows are direction strata, not 774 distinct MLG types.

Source. J.-X. Du, S. Zeng and Y.-J. Zhao, *Stacking theory for bilayer two-dimensional magnets*, Phys. Rev. B **113**, 214455 (2026), DOI: 10.1103/g471-xrkl; Supplemental Material, Table S7, pp. 96–113. The source supplies the cSLGs and operations. The Néel-direction branches, SOC MLGs and response constraints in this report are calculated results.

Direction notation. $\mathrm{AX}_{[uvw]}$ denotes a special axis; $\mathrm{PL}_{(hkl)}$ denotes a direction in a special plane excluding all axes lying in that plane; GEN denotes a direction outside every listed axis and plane. Only symmetry-distinct stabilizer strata of a given group are introduced; for example, the first group needs only $\mathrm{AX}_{[100]}$, $\mathrm{PL}_{(100)}$ and GEN.

Coordinates. Tensor components are Cartesian, with z normal to the layer. For rectangular and tetragonal groups, $x \parallel \mathbf{a}$, $y \parallel \mathbf{b}$. For hexagonal groups, $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1/2, \sqrt{3}/2, 0)$, while Cartesian $y \parallel \hat{z} \times \mathbf{a}$. Miller indices refer to these crystallographic bases. Thus [010] is not the Cartesian y direction in the hexagonal setting. Hexagonal in-plane operations are displayed with explicit crystallographic directions; $M_{[uvw]}$ has its Cartesian normal along $[uvw]$. The source triplets $\{x, y, xy\}$ and $\{1, 2, 3\}$ correspond as sets to $\{[100], [010], [110]\}$ and $\{[1\bar{1}0], [120], [210]\}$. No individual identification of the source's numbered subscripts is needed.

Operation columns. Translation parts are omitted and repeated point operations are merged. All distinct nontrivial-part SLG point operations are shown on separate lines; the common spin-only factor is implicit. Every MLG row explicitly lists its point operations on one line. Full MLG symbols and numbers are nevertheless identified from the full affine operations in the source, then verified by a basis/origin change preserving the layer normal. MLG symbols use their standard settings; the explicitly listed operations and tensor indices remain in the source coordinate setting. An affine witness for every row is included in the data package. Omitting translations from the displayed columns does not redefine these group identifiers.

Tensor convention. Set $A^\ell = \chi^\ell_{xyy}$ and $B^\ell = \chi^\ell_{yxx}$ for $\ell = t, b$, with $\chi_{abc} = -\chi_{bac}$ and in-plane indices. Then $\chi_{xyx} = -B$, $\chi_{yxy} = -A$, and all other in-plane entries vanish. The two-component vector $\mathbf{v}^\ell = (A^\ell, B^\ell)$ transforms as $\mathbf{v}^{g\ell} = q_g R_\parallel \mathbf{v}^\ell$. This statement is specific to the $\mathcal{T}$-odd intrinsic Hall part.

Mark	Constraint on the displayed component	Meaning
$+$	$\chi^t = \chi^b$, allowed	Equal layer contributions
$-$	$\chi^t = -\chi^b$, allowed	Hidden, layer compensated
0	$\chi^t = \chi^b = 0$	Forbidden in each layer
v	No fixed proportional relation between the same component in the two layers	See coupled constraints when starred

A star on a direction indicates additional relations between A and B, listed completely in Appendix A (208 branches). In these rows, v must not be read as absence of all symmetry relations: cross-component relations may remain. All allowed values can accidentally vanish microscopically.

Assumptions and calculation. Fixed structure and collinear order; two physical layers at opposite sides of $z = 0$. The SOC subgroup satisfies $q_g \det(R_g) R_g \hat{\mathbf{n}} = s_g \hat{\mathbf{n}}$, with $q_g = +1$ for g and $q_g = -1$ for $g\mathcal{T}$. Operations with $R_{zz} = -1$ exchange the two layers. Branches are constructed from all real ± 1 eigenspaces and their intersections. These are symmetry-allowed constrained orientations, not predictions of magnetic easy axes or stability.

Source audit. All 92 target operation lists close as printed, and their spin labels agree with the printed cSLG symbols; no target operation was modified. The reference catalog used for MLG identification retains three explicitly documented consistency repairs: 31.5.192, operation 04, $s_{M_x} = +1$; 39.5.260, operation 08, translation $(0, 1/2, 0)$; and FM reference 77.1.505, operation 09, $s_{M_y} = +1$. These are consistency choices derived from group closure, not an author-issued erratum. Original source strings are retained in the data package.

Validation. All 774 MLG assignments are checked against complete affine operations. Each allowed response space is independently checked using the full third-rank Hall tensor, and direction coverage is sampled on every axis and plane as well as generic orientations.

Complete table, in Table S7 order

First three columns are merged across all direction branches of one cSLG. Asterisks refer to Appendix A.

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
01 8.3.36 $p^{-1}2^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$	$AX_{[100]}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$PL_{(100)}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	$+$	$-$
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
02 9.3.43 $p^{-1}2_1^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$	$AX_{[100]}$	$p2_1'11$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$PL_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
03 10.3.47 $c^{-1}2^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$	$AX_{[100]}$	$c2'11$ 10.3.47	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$PL_{(100)}$	$c211$ 10.1.45	$\{E, C_{2x}\}$	$+$	$-$
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
04 11.3.52 $p^{-1}m^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
05 12.3.59 $p^{-1}b^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			$PL_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
06 13.3.63 $c^{-1}m^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$cm'11$ 13.3.63	$\{E, M_x\mathcal{T}\}$	v	0
			$PL_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
07 14.5.70 $p^{-1}2/^{-1}m^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	$-$	0
			$PL_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	$-$
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	$-$	$-$
08 15.4.79 $p^{-1}2_1/^{-1}m^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p2_1'/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	$-$	0
			$PL_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	$-$
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	$-$	$-$

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
09 16.4.86 $p^{-1}2/^{-1}b^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(100)}$	$p\bar{2}/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
10 17.4.93 $p^{-1}2_1/^{-1}b^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2'_1/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
11 18.4.98 $c^{-1}2/^{-1}m^11^11$ SI p. 96	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}_{(100)}$	$c2/m11$ 18.1.95	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
12 19.3.106 $p^{-1}2^{-1}2^12$ SI p. 96	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	–	+
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	–
			$\mathrm{AX}_{[001]}$	$p222$ 19.1.104	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p\bar{2}11$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$\mathrm{PL}_{(100)}$	$p\bar{2}11$ 8.1.34	$\{E, C_{2x}\}$	+	–
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
13 20.5.115 $p^{-1}2_1^{-1}2^12$ SI p. 97	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$	$\mathrm{AX}_{[100]}$	$p2'_122'$ 20.4.114	$\{E, C_{2x}\mathcal{T}, C_{2y}, C_{2z}\mathcal{T}\}$	–	+
			$\mathrm{AX}_{[010]}$	$p2_12'2'$ 20.3.113	$\{E, C_{2x}, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}\}$	+	–
			$\mathrm{AX}_{[001]}$	$p2_122$ 20.1.111	$\{E, C_{2x}, C_{2y}, C_{2z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p\bar{2}11$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	+	–
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
14 21.3.120 $p^{-1}2_1^{-1}2_1^12$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[001]}$	$p2_12_12$ 21.1.118	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
15 22.3.124 $c^{-1}2^{-1}2^12$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$	$\mathrm{AX}_{[100]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[001]}$	$c222$ 22.1.122	$\{E, C_{2z}, C_{2y}, C_{2x}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$c211$ 10.1.45	$\{E, C_{2x}\}$	$+$	$-$
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
16 23.3.131 $p^{-1}m^{-1}m^12$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[001]}$	$pmm2$ 23.1.129	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
17 24.3.138 $p^{-1}m^{-1}a^12$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'a2'$ 24.4.139	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pma'2'$ 24.5.140	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[001]}$	$pma2$ 24.1.136	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
18 25.3.145 $p^{-1}b^{-1}a^{1}2$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[001]}$	$pba2$ 25.1.143	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
19 26.3.149 $c^{-1}m^{-1}m^{1}2$ SI p. 97	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[001]}$	$cmm2$ 26.1.147	$\{E, C_{2z}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
20 27.5.158 $p^{-1}m^{-1}2^{1}m$ SI p. 97	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pm'2m'$ 27.3.156	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pm2'm'$ 27.4.157	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[001]}$	$pm2m$ 27.1.154	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
21 28.4.170 $p^{-1}m^{-1}2_1^1b$ SI p. 97	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'2_1b'$ 28.3.169	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pm2_1'b'$ 28.5.171	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pm2_1b$ 28.1.167	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$\mathrm{PL}_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	–	+
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
22 29.5.178 $p^{-1}b^{-1}2_1^1m$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$pb'2_1m'$ 29.3.176	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pb2_1'm'$ 29.4.177	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[001]}$	$pb2_1m$ 29.1.174	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$PL_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
23 30.4.184 $p^{-1}b^{-1}2^1b$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pb'2b'$ 30.3.183	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pb2'b'$ 30.5.185	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$AX_{[001]}$	$pb2b$ 30.1.181	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
24 31.4.191 $p^{-1}m^{-1}2^1a$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'2a'$ 31.3.190	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pm2'a'$ 31.5.192	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$AX_{[001]}$	$pm2a$ 31.1.188	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
25 32.4.200 $p^{-1}m^{-1}2_1^1n$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pm'2_1n'$ 32.3.199	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pm2_1'n'$ 32.5.201	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$AX_{[001]}$	$pm2_1n$ 32.1.197	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
26 33.4.205 $p^{-1}b^{-1}2^1_1a$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pb'2_1a'$ 33.3.204	$\{E, C_{2y}, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pb2'_1a'$ 33.5.206	$\{E, C_{2y}\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$AX_{[001]}$	$pb2_1a$ 33.1.202	$\{E, C_{2y}, M_z, M_x\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
27 34.5.211 $p^{-1}b^{-1}2^1n$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$pb'2n'$ 34.3.209	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[010]}$	$pb2'n'$ 34.4.210	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[001]}$	$pb2n$ 34.1.207	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
28 35.5.216 $c^{-1}m^{-1}2^1m$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$cm'2m'$ 35.3.214	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[010]}$	$cm2'm'$ 35.4.215	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[001]}$	$cm2m$ 35.1.212	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$PL_{(001)}$	$p11m'$ 4.3.14	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
29 36.5.225 $c^{-1}m^{-1}2^1a$ SI p. 98	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$AX_{[100]}$	$cm'2a'$ 36.3.223	$\{E, C_{2y}, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$AX_{[010]}$	$cm2'a'$ 36.4.224	$\{E, C_{2y}\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$AX_{[001]}$	$cm2a$ 36.1.221	$\{E, C_{2y}, M_x, M_z\}$	0	+
			$PL_{(001)}$	$p11a'$ 5.3.19	$\{E, M_z\mathcal{T}\}$	–	–
			$PL_{(010)}$	$c211$ 10.1.45	$\{E, C_{2y}\}$	–	+
			$PL_{(100)}$	$cm11$ 13.1.61	$\{E, M_x\}$	0	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
30 37.4.233 $p^{-1}m^{-1}m^1m$ SI p. 98	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pmmm$ 37.1.230	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
31 38.5.247 $p^{-1}m^{-1}a^1a$ SI p. 99	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pm'aa'$ 38.6.248	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	–	0
			$\mathrm{AX}_{[010]}$	$pma'a'$ 38.4.246	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[001]}$	$pmaa$ 38.1.243	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
32 39.4.259 $p^{-1}b^{-1}a^1n$ SI p. 99	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pban$ 39.1.256	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
33 40.5.267 $p^{-1}m^{-1}a^1m$ SI p. 99	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'am'$ 40.4.266	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pma'm'$ 40.6.268	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pmam$ 40.1.263	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
34 41.4.279 $p^{-1}m^{-1}m^{1}a$ SI p. 99	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'ma'$ 41.5.280	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'a'$ 41.6.281	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pmma$ 41.1.276	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
35 42.5.293 $p^{-1}m^{-1}a^{1}n$ SI p. 99	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'an'$ 42.4.292	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}, M_y, M_z\mathcal{T}, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pma'n'$ 42.6.294	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pman$ 42.1.289	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}, M_y, M_z, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
36 43.6.303 $p^{-1}b^{-1}a^{1}a$ SI p. 100	AM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pb'aa'$ 43.5.302	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'a'$ 43.4.301	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}, M_x, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[001]}$	$pbaa$ 43.1.298	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}, M_x, M_z, M_y\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
37 44.4.310 $p^{-1}b^{-1}a^{1}m$ SI p. 100	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pbam$ 44.1.307	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
38 45.6.319 $p^{-1}b^{-1}m^1a$ SI p. 100	AM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'ma'$ 45.4.317	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}, M_y, M_x\mathcal{T}, M_z\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pbm'a'$ 45.5.318	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}, M_x, M_z\mathcal{T}\}$	0	–
			$\mathrm{AX}_{[001]}$	$pbma$ 45.1.314	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}, M_y, M_x, M_z\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
39 46.4.326 $p^{-1}m^{-1}m^1n$ SI p. 100	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$pmmn$ 46.1.323	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
40 47.4.333 $c^{-1}m^{-1}m^1m$ SI p. 100	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$cmmm$ 47.1.330	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$c2/m11$ 18.1.95	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
41 48.4.346 $c^{-1}m^{-1}m^1e$ SI p. 101	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[001]}$	$cmme$ 48.1.343	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}, M_z, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$c2/m11$ 18.1.95	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
42 49.3.356 $p^{-1}4$ SI p. 101	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4'$ 49.3.356	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
43 50.3.360 $p^{-1}\bar{4}$ SI p. 101	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p\bar{4}'$ 50.3.360	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			GEN	$p\bar{1}$ 1.1.1	$\{E\}$	v	v
44 51.5.366 $p^{-1}4/^1m$ SI p. 101	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4'/m$ 51.5.366	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
45 52.5.373 $p^{-1}4/^1n$ SI p. 101	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4'/n$ 52.5.373	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
46 53.3.376 $p^14^{-1}2^{-1}2$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	–	+
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	+	–
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p422$ 53.1.374	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	–	+
			$\mathrm{PL}_{(100)}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	+	–
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
47 53.4.377 $p^{-1}4^12^{-1}2$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'2'2$ 53.5.378	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
48 53.5.378 $p^{-1}4^{-1}2^12$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'22'$ 53.4.377	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
49 54.3.383 $p^14^{-1}2_1^{-1}2$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p42_12$ 54.1.381	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_11\bar{1}$ 9.1.41	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
50 54.4.384 $p^{-1}4^12_1^{-1}2$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'2'_12$ 54.5.385	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'_111$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2'_111$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
51 54.5.385 $p^{-1}4^{-1}2_1^12$ SI p. 102	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'2_12'$ 54.4.384	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
52 55.3.388 $p^14^{-1}m^{-1}m$ SI p. 103	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4mm$ 55.1.386	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
53 55.4.389 $p^{-1}4^1m^{-1}m$ SI p. 103	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'm'm$ 55.5.390	$\{E, C_{2z}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm'11$ 11.3.52	$\{E, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
54 55.5.390 $p^{-1}4^{-1}m^1m$ SI p. 103	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'mm'$ 55.4.389	$\{E, C_{2z}, C_{4z}\mathcal{T}, C^{-1}_{4z}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
55 56.3.395 $p^14^{-1}b^{-1}m$ SI p. 103	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4bm$ 56.1.393	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
56 56.4.396 $p^{-1}4^1b^{-1}m$ SI p. 103	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'b'm$ 56.5.397	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb'11$ 12.3.59	$\{E, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
57 56.5.397 $p^{-1}4^{-1}b^1m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p4'bm'$ 56.4.396	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
58 57.3.400 $p^1\bar{4}^{-1}2^{-1}m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}2m$ 57.1.398	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, C_{2y}, C_{2x}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
59 57.4.401 $p^{-1}\bar{4}^{1}2^{-1}m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^{*}_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'2'm$ 57.5.402	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'11$ 8.3.36	$\{E, C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2'11$ 8.3.36	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^{*}_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
60 57.5.402 $p^{-1}\bar{4}^{-1}2^{1}m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p22'2'$ 19.4.107	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^{*}_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'2m'$ 57.4.401	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, C_{2y}, C_{2x}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p211$ 8.1.34	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p211$ 8.1.34	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^{*}_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
61 58.3.407 $p^1\bar{4}^{-1}2_1^{-1}m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}2_1m$ 58.1.405	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, C_{2y}, C_{2x}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_11$ 9.1.41	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p2_11$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
62 58.4.408 $p^{-1}\bar{4}^12_1^{-1}m$ SI p. 104	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}_{[010]}$	$p2_12'_12'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'2'_1m$ 58.5.409	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2'_11$ 9.3.43	$\{E, C_{2y}\mathcal{T}\}$	$+$	$-$
			$\mathrm{PL}_{(100)}$	$p2'_11$ 9.3.43	$\{E, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
63 58.5.409 $p^{-1}\bar{4}^{-1}2_1^1m$ SI p. 105	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}\}$	$-$	$+$
			$\mathrm{AX}_{[010]}$	$p2_12_1'2'$ 21.4.121	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\}$	$+$	$-$
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'2_1m'$ 58.4.408	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$p2_111$ 9.1.41	$\{E, C_{2y}\}$	$-$	$+$
			$\mathrm{PL}_{(100)}$	$p2_111$ 9.1.41	$\{E, C_{2x}\}$	$+$	$-$
			$\mathrm{PL}^*_{(110)}$	$cm'11$ 13.3.63	$\{E, M_{xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm'11$ 13.3.63	$\{E, M_{\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
64 59.3.412 $p^1\bar{4}^{-1}m^{-1}2$ SI p. 105	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}m2$ 59.1.410	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
65 59.4.413 $p^{-1}\bar{4}^{1}m^{-1}2$ SI p. 105	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^{*}_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'m'2$ 59.5.414	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm'11$ 11.3.52	$\{E, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pm'11$ 11.3.52	$\{E, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^{*}_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
66 59.5.414 $p^{-1}\bar{4}^{-1}m^{1}2$ SI p. 105	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pmm'2'$ 23.4.132	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^{*}_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'m2'$ 59.4.413	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y, M_x, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pm11$ 11.1.50	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pm11$ 11.1.50	$\{E, M_x\}$	0	v
			$\mathrm{PL}^{*}_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
67 60.3.419 $p^1\bar{4}^{-1}b^{-1}2$ SI p. 105	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}b2$ 60.1.417	$\{E, C_{2z}, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y, M_x, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
68 60.4.420 $p^{-1}\bar{4}^1b^{-1}2$ SI p. 106	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'b'2$ 60.5.421	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb'11$ 12.3.59	$\{E, M_y\mathcal{T}\}$	0	v
			$\mathrm{PL}_{(100)}$	$pb'11$ 12.3.59	$\{E, M_x\mathcal{T}\}$	v	0
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2xy}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2\bar{x}y}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
69 60.5.421 $p^{-1}\bar{4}^{-1}b^{1}2$ SI p. 106	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y, M_x\mathcal{T}\}$	v	0
			$\mathrm{AX}_{[010]}$	$pba'2'$ 25.4.146	$\{E, C_{2z}\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	v
			$\mathrm{AX}^{*}_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{4}'b2'$ 60.4.420	$\{E, C_{2z}, \bar{C}_{4z}\mathcal{T}, \bar{C}^{-1}_{4z}\mathcal{T}, M_y, M_x, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11'2'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$pb11$ 12.1.57	$\{E, M_y\}$	v	0
			$\mathrm{PL}_{(100)}$	$pb11$ 12.1.57	$\{E, M_x\}$	0	v
			$\mathrm{PL}^{*}_{(110)}$	$c2'11$ 10.3.47	$\{E, C_{2xy}\mathcal{T}\}$	v	v
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2'11$ 10.3.47	$\{E, C_{2\bar{x}y}\mathcal{T}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
70 61.4.425 $p^{1}4/^{1}m^{-1}m^{-1}m$ SI p. 106	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C^{-1}_{4z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}^{-1}_{4z}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^{*}_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/mmm$ 61.1.422	$\{E, C_{2z}, C_{4z}, C^{-1}_{4z}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}^{-1}_{4z}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^{*}_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
71 61.8.429 $p^{-1}4/^1m^1m^{-1}m$ SI p. 106	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/mm'm$ 61.9.430	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/m'11$ 14.5.70	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
72 61.9.430 $p^{-1}4/^1m^{-1}m^1m$ SI p. 107	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'm'$ 37.5.234	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/mmm'$ 61.8.429	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/m11$ 14.1.66	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/m11$ 14.1.66	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
73 62.4.438 $p^14/^1n^{-1}b^{-1}m$ SI p. 107	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/nbm$ 62.1.435	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
74 62.8.442 $p^{-1}4/^1n^1b^{-1}m$ SI p. 107	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/nb'm$ 62.9.443	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2'/b'11$ 16.4.86	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
75 62.9.443 $p^{-1}4/^1n^{-1}b^1m$ SI p. 108	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'n'$ 39.5.260	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/nbm'$ 62.8.442	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2/b11$ 16.1.83	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2/b11$ 16.1.83	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
76 63.4.447 $p^14/^1m^{-1}b^{-1}m$ SI p. 108	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/mbm$ 63.1.444	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
77 63.8.451 $p^{-1}4/^1m^1b^{-1}m$ SI p. 108	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/mb'm$ 63.9.452	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1'/b'11$ 17.4.93	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1'/b'11$ 17.4.93	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
78 63.9.452 $p^{-1}4/^1m^{-1}b^1m$ SI p. 109	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pba'm'$ 44.5.311	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/mbm'$ 63.8.451	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/b11$ 17.1.90	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
79 64.4.456 $p^14/^1n^{-1}m^{-1}m$ SI p. 109	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4/nmm$ 64.1.453	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
80 64.8.460 $p^{-1}4/^1n^1m^{-1}m$ SI p. 110	AM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/nm'm$ 64.9.461	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1'/m'11$ 15.4.79	$\{E, C_{2y}\mathcal{T}, \mathcal{P}, M_y\mathcal{T}\}$	0	–
			$\mathrm{PL}_{(100)}$	$p2_1'/m'11$ 15.4.79	$\{E, C_{2x}\mathcal{T}, \mathcal{P}, M_x\mathcal{T}\}$	–	0
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2xy}, \mathcal{P}, M_{xy}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2\bar{x}y}, \mathcal{P}, M_{\bar{x}y}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
81 64.9.461 $p^{-1}4/^1n^{-1}m^1m$ SI p. 110	AM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{-1\|C_{4z}\}$, $\{-1\|C_{4z}^{-1}\}$, $\{-1\|C_{2y}\}$, $\{-1\|C_{2x}\}$, $\{1\|C_{2xy}\}$, $\{1\|C_{2\bar{x}y}\}$, $\{1\|\mathcal{P}\}$, $\{1\|M_z\}$, $\{-1\|\bar{C}_{4z}\}$, $\{-1\|\bar{C}_{4z}^{-1}\}$, $\{-1\|M_y\}$, $\{-1\|M_x\}$, $\{1\|M_{xy}\}$, $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_y, M_x\mathcal{T}\}$	–	0
			$\mathrm{AX}_{[010]}$	$pmm'n'$ 46.5.327	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\}$	0	–
			$\mathrm{AX}^*_{[110]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e'$ 48.5.347	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}, M_z\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p4'/nmm'$ 64.8.460	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_z, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a'$ 7.5.32	$\{E, C_{2z}\mathcal{T}, \mathcal{P}, M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2y}, \mathcal{P}, M_y\}$	–	0
			$\mathrm{PL}_{(100)}$	$p2_1/m11$ 15.1.76	$\{E, C_{2x}, \mathcal{P}, M_x\}$	0	–
			$\mathrm{PL}^*_{(110)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}, M_{xy}\mathcal{T}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m'11$ 18.4.98	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}, M_{\bar{x}y}\mathcal{T}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
82 67.3.469 $p^13^11^{-1}2$ SI p. 110	AM	$\{1\|E\}$, $\{1\|C_{3z}\}$, $\{1\|C_{3z}^{-1}\}$, $\{-1\|C_{2,[210]}\}$, $\{-1\|C_{2,[120]}\}$, $\{-1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$c2'11$ 10.3.47	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[120]}$	$c2'11$ 10.3.47	$\{E, C_{2,[120]}\mathcal{T}\}$	+	–
			$\mathrm{AX}^*_{[210]}$	$c2'11$ 10.3.47	$\{E, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{3}12$ 67.1.467	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2,[120]}\}$	–	+
			$\mathrm{PL}^*_{(100)}$	$c211$ 10.1.45	$\{E, C_{2,[210]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[1\bar{1}0]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
83 68.3.472 $p^13^{-1}2^11$ SI p. 110	AM	$\{1\|E\}$, $\{1\|C_{3z}\}$, $\{1\|C_{3z}^{-1}\}$, $\{-1\|C_{2,[110]}\}$, $\{-1\|C_{2,[100]}\}$, $\{-1\|C_{2,[010]}\}$	$\mathrm{AX}_{[100]}$	$c2'11$ 10.3.47	$\{E, C_{2,[100]}\mathcal{T}\}$	–	+
			$\mathrm{AX}^*_{[010]}$	$c2'11$ 10.3.47	$\{E, C_{2,[010]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$c2'11$ 10.3.47	$\{E, C_{2,[110]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p321$ 68.1.470	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2,[110]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[100]}\}$	+	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c211$ 10.1.45	$\{E, C_{2,[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
84 69.3.475 $p^13^{-1}m^11$ SI p. 111	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$cm'11$ 13.3.63	$\{E, M_{[100]}\mathcal{T}\}$	v	0
			$\mathrm{AX}^*_{[010]}$	$cm'11$ 13.3.63	$\{E, M_{[010]}\mathcal{T}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$cm'11$ 13.3.63	$\{E, M_{[110]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{3}m1$ 69.1.473	$\{E, C_{3z}, C_{3z}^{-1}, M_{[110]}, M_{[100]}, M_{[010]}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{[110]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[100]}\}$	0	v
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm11$ 13.1.61	$\{E, M_{[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v
85 70.3.478 $p^13^11^{-1}m$ SI p. 111	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm'11$ 13.3.63	$\{E, M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[120]}$	$cm'11$ 13.3.63	$\{E, M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{AX}^*_{[210]}$	$cm'11$ 13.3.63	$\{E, M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{3}1m$ 70.1.476	$\{E, C_{3z}, C_{3z}^{-1}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E, M_{[120]}\}$	v	0
			$\mathrm{PL}^*_{(100)}$	$cm11$ 13.1.61	$\{E, M_{[210]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[1\bar{1}0]}\}$	v	v
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	v	v
86 71.4.482 $p^1\bar{3}^11^{-1}m$ SI p. 111	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^*_{[1\bar{1}0]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}, M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[120]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}, M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{AX}^*_{[210]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}, M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{3}1m$ 71.1.479	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[120]}, \mathcal{P}, M_{[120]}\}$	–	0
			$\mathrm{PL}^*_{(100)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[210]}, \mathcal{P}, M_{[210]}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[1\bar{1}0]}, \mathcal{P}, M_{[1\bar{1}0]}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
87 72.4.487 $p^1\bar{3}^{-1}m^11$ SI p. 111	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{P}, M_{[100]}\mathcal{T}\}$	–	0
			$\mathrm{AX}^*_{[010]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{P}, M_{[010]}\mathcal{T}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$c2'/m'11$ 18.4.98	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{P}, M_{[110]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{3}m1$ 72.1.484	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, \mathcal{P}, \bar{C}_{3z}, \bar{C}_{3z}^{-1}, M_{[110]}, M_{[100]}, M_{[010]}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[110]}, \mathcal{P}, M_{[110]}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[100]}, \mathcal{P}, M_{[100]}\}$	0	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m11$ 18.1.95	$\{E, C_{2,[010]}, \mathcal{P}, M_{[010]}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E, \mathcal{P}\}$	–	–
88 76.3.502 $p^16^{-1}2^{-1}2$ SI p. 111	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\}$	–	+
			$\mathrm{AX}^*_{[010]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}, C_{2,[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[120]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}, C_{2,[120]}\mathcal{T}\}$	+	–
			$\mathrm{AX}^*_{[210]}$	$c22'2'$ 22.4.125	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}, C_{2,[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p\bar{6}22$ 76.1.500	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11\bar{2}'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E, C_{2,[120]}\}$	–	+
			$\mathrm{PL}^*_{(100)}$	$c211$ 10.1.45	$\{E, C_{2,[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E, C_{2,[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c211$ 10.1.45	$\{E, C_{2,[100]}\}$	+	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c211$ 10.1.45	$\{E, C_{2,[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
89 77.3.507 $p^16^{-1}m^{-1}m$ SI p. 112	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[120]}\}$	v	0
			$\mathrm{AX}^*_{[010]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}\}$	v	v
			$\mathrm{AX}^*_{[110]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[110]}, M_{[1\bar{1}0]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[120]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[100]}, M_{[120]}\mathcal{T}\}$	0	v
			$\mathrm{AX}^*_{[210]}$	$cmm'2'$ 26.4.150	$\{E, C_{2z}\mathcal{T}, M_{[010]}, M_{[210]}\mathcal{T}\}$	v	v
			$\mathrm{AX}_{[001]}$	$p6mm$ 77.1.505	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}, M_{[110]}, M_{[100]}, M_{[010]}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'$ 3.3.10	$\{E, C_{2z}\mathcal{T}\}$	v	v
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E, M_{[120]}\}$	v	0
			$\mathrm{PL}^*_{(100)}$	$cm11$ 13.1.61	$\{E, M_{[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E, M_{[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm11$ 13.1.61	$\{E, M_{[100]}\}$	0	v
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm11$ 13.1.61	$\{E, M_{[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
90 78.3.512 $p^1\bar{6}^{-1}m^{-1}2$ SI p. 112	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[100]}\mathcal{T},\ C_{2,[120]}\}$	–	0
			$\mathrm{AX}^*_{[010]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[010]}\mathcal{T},\ C_{2,[210]}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ M_{[110]}\mathcal{T},\ C_{2,[1\bar{1}0]}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[110]},\ C_{2,[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[120]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[100]},\ C_{2,[120]}\mathcal{T}\}$	0	–
			$\mathrm{AX}^*_{[210]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ M_{[010]},\ C_{2,[210]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{6}m2$ 78.1.510	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z,\ \bar{C}_{6z}^{-1},\ \bar{C}_{6z},\ M_{[110]},\ M_{[100]},\ M_{[010]},\ C_{2,[210]},\ C_{2,[120]},\ C_{2,[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E,\ M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c211$ 10.1.45	$\{E,\ C_{2,[120]}\}$	–	+
			$\mathrm{PL}^*_{(100)}$	$c211$ 10.1.45	$\{E,\ C_{2,[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$cm11$ 13.1.61	$\{E,\ M_{[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c211$ 10.1.45	$\{E,\ C_{2,[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[100]}\}$	0	v
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
91 79.3.517 $p^1\bar{6}^{-1}2^{-1}m$ SI p. 112	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[100]}\mathcal{T},\ M_{[120]}\}$	–	0
			$\mathrm{AX}^*_{[010]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[010]}\mathcal{T},\ M_{[210]}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$cm2'm'$ 35.4.215	$\{E,\ M_z\mathcal{T},\ C_{2,[110]}\mathcal{T},\ M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[110]},\ M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[120]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[100]},\ M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{AX}^*_{[210]}$	$cm'2m'$ 35.3.214	$\{E,\ M_z\mathcal{T},\ C_{2,[010]},\ M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p\bar{6}2m$ 79.1.515	$\{E,\ C_{3z},\ C_{3z}^{-1},\ M_z,\ \bar{C}_{6z}^{-1},\ \bar{C}_{6z},\ C_{2,[110]},\ C_{2,[100]},\ C_{2,[010]},\ M_{[210]},\ M_{[120]},\ M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11m'$ 4.3.14	$\{E,\ M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$cm11$ 13.1.61	$\{E,\ M_{[120]}\}$	v	0
			$\mathrm{PL}^*_{(100)}$	$cm11$ 13.1.61	$\{E,\ M_{[210]}\}$	v	v
			$\mathrm{PL}^*_{(110)}$	$c211$ 10.1.45	$\{E,\ C_{2,[110]}\}$	v	v
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$cm11$ 13.1.61	$\{E,\ M_{[1\bar{1}0]}\}$	v	v
			$\mathrm{PL}_{(2\bar{1}0)}$	$c211$ 10.1.45	$\{E,\ C_{2,[100]}\}$	+	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c211$ 10.1.45	$\{E,\ C_{2,[010]}\}$	v	v
			GEN	$p1$ 1.1.1	$\{E\}$	v	v

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
92 80.4.523 $p^16/^1m^{-1}m^{-1}m$ SI p. 112	AM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{1\|\mathcal{P}\}$ $\{1\|\bar{C}_{3z}\}$ $\{1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[100]}\mathcal{T},\ C_{2,[120]},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[100]}\mathcal{T},\ M_{[120]}\}$	–	0
			$\mathrm{AX}^*_{[010]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[010]}\mathcal{T},\ C_{2,[210]},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[010]}\mathcal{T},\ M_{[210]}\}$	–	–
			$\mathrm{AX}^*_{[110]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[110]}\mathcal{T},\ C_{2,[1\bar{1}0]},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[110]}\mathcal{T},\ M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[110]},\ C_{2,[1\bar{1}0]}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[110]},\ M_{[1\bar{1}0]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[120]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[100]},\ C_{2,[120]}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[100]},\ M_{[120]}\mathcal{T}\}$	0	–
			$\mathrm{AX}^*_{[210]}$	$cmm'm'$ 47.5.334	$\{E,\ C_{2z}\mathcal{T},\ C_{2,[010]},\ C_{2,[210]}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T},\ M_{[010]},\ M_{[210]}\mathcal{T}\}$	–	–
			$\mathrm{AX}_{[001]}$	$p6/mmm$ 80.1.520	$\{E,\ C_{3z},\ C_{3z}^{-1},\ C_{2z},\ C_{6z}^{-1},\ C_{6z},\ C_{2,[110]},\ C_{2,[100]},\ C_{2,[010]},\ C_{2,[210]},\ C_{2,[120]},\ C_{2,[1\bar{1}0]},\ \mathcal{P},\ \bar{C}_{3z},\ \bar{C}_{3z}^{-1},\ M_z,\ \bar{C}_{6z}^{-1},\ \bar{C}_{6z},\ M_{[110]},\ M_{[100]},\ M_{[010]},\ M_{[210]},\ M_{[120]},\ M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m'$ 6.5.25	$\{E,\ C_{2z}\mathcal{T},\ \mathcal{P},\ M_z\mathcal{T}\}$	–	–
			$\mathrm{PL}_{(010)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[120]},\ \mathcal{P},\ M_{[120]}\}$	–	0
			$\mathrm{PL}^*_{(100)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[210]},\ \mathcal{P},\ M_{[210]}\}$	–	–
			$\mathrm{PL}^*_{(110)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[110]},\ \mathcal{P},\ M_{[110]}\}$	–	–
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[1\bar{1}0]},\ \mathcal{P},\ M_{[1\bar{1}0]}\}$	–	–
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[100]},\ \mathcal{P},\ M_{[100]}\}$	0	–
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m11$ 18.1.95	$\{E,\ C_{2,[010]},\ \mathcal{P},\ M_{[010]}\}$	–	–
			GEN	$p\bar{1}$ 2.1.4	$\{E,\ \mathcal{P}\}$	–	–

Appendix A. Complete coupled-component constraints

$A^\ell = \chi^\ell_{xyy}$, $B^\ell = \chi^\ell_{yxx}$. Each row below gives the entire allowed linear space for a starred branch, with independent real parameters u or u_1, u_2. These relations supplement the $+, -, 0, v$ flags; they are not additional assumptions.

No.	cSLG label	Néel direction	Complete allowed space
46	53.3.376	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
46	53.3.376	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
46	53.3.376	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
46	53.3.376	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
47	53.4.377	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
47	53.4.377	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
47	53.4.377	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
47	53.4.377	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
48	53.5.378	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
48	53.5.378	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
48	53.5.378	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
48	53.5.378	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
49	54.3.383	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
49	54.3.383	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
49	54.3.383	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
49	54.3.383	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
50	54.4.384	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
50	54.4.384	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
50	54.4.384	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
50	54.4.384	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
51	54.5.385	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
51	54.5.385	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
51	54.5.385	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(0, -1, 1, 0) + u_2(-1, 0, 0, 1)$
51	54.5.385	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(0, 1, 1, 0) + u_2(1, 0, 0, 1)$
52	55.3.388	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
52	55.3.388	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
52	55.3.388	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
52	55.3.388	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
53	55.4.389	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
53	55.4.389	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
53	55.4.389	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
53	55.4.389	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
54	55.5.390	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
54	55.5.390	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
54	55.5.390	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
54	55.5.390	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
55	56.3.395	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
55	56.3.395	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
55	56.3.395	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
55	56.3.395	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
56	56.4.396	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
56	56.4.396	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
56	56.4.396	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
56	56.4.396	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$
57	56.5.397	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1(-1, 1, 0, 0) + u_2(0, 0, -1, 1)$
57	56.5.397	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1(1, 1, 0, 0) + u_2(0, 0, 1, 1)$

No.	cSLG label	Néel direction	Complete allowed space
57	56.5.397	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
57	56.5.397	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
58	57.3.400	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
58	57.3.400	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
58	57.3.400	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
58	57.3.400	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
59	57.4.401	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
59	57.4.401	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
59	57.4.401	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
59	57.4.401	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
60	57.5.402	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
60	57.5.402	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
60	57.5.402	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
60	57.5.402	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
61	58.3.407	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
61	58.3.407	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
61	58.3.407	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
61	58.3.407	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
62	58.4.408	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
62	58.4.408	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
62	58.4.408	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
62	58.4.408	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
63	58.5.409	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
63	58.5.409	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
63	58.5.409	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,1,\,1\big)$
63	58.5.409	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\,1,\,0,\,0\big) + u_2\big(0,\,0,\,-1,\,1\big)$
64	59.3.412	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
64	59.3.412	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
64	59.3.412	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
64	59.3.412	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
65	59.4.413	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
65	59.4.413	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
65	59.4.413	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
65	59.4.413	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
66	59.5.414	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
66	59.5.414	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
66	59.5.414	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
66	59.5.414	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
67	60.3.419	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
67	60.3.419	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
67	60.3.419	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
67	60.3.419	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
68	60.4.420	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
68	60.4.420	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
68	60.4.420	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
68	60.4.420	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
69	60.5.421	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
69	60.5.421	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$
69	60.5.421	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,-1,\,1,\,0\big) + u_2\big(-1,\,0,\,0,\,1\big)$

No.	cSLG label	Néel direction	Complete allowed space
69	60.5.421	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(0,\,1,\,1,\,0\big) + u_2\big(1,\,0,\,0,\,1\big)$
70	61.4.425	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
70	61.4.425	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
70	61.4.425	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
70	61.4.425	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
71	61.8.429	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
71	61.8.429	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
71	61.8.429	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
71	61.8.429	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
72	61.9.430	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
72	61.9.430	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
72	61.9.430	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
72	61.9.430	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
73	62.4.438	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
73	62.4.438	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
73	62.4.438	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
73	62.4.438	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
74	62.8.442	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
74	62.8.442	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
74	62.8.442	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
74	62.8.442	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
75	62.9.443	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
75	62.9.443	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
75	62.9.443	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
75	62.9.443	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
76	63.4.447	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
76	63.4.447	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
76	63.4.447	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
76	63.4.447	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
77	63.8.451	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
77	63.8.451	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
77	63.8.451	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
77	63.8.451	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
78	63.9.452	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
78	63.9.452	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
78	63.9.452	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
78	63.9.452	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
79	64.4.456	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
79	64.4.456	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
79	64.4.456	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
79	64.4.456	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
80	64.8.460	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
80	64.8.460	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
80	64.8.460	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
80	64.8.460	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
81	64.9.461	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$
81	64.9.461	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
81	64.9.461	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\,-1,\,1,\,1\big)$
81	64.9.461	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\,-1,\,-1,\,1\big)$

No.	cSLG label	Néel direction	Complete allowed space
82	67.3.469	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
82	67.3.469	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
82	67.3.469	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
82	67.3.469	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
83	68.3.472	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
83	68.3.472	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
83	68.3.472	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
83	68.3.472	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
84	69.3.475	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
84	69.3.475	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
84	69.3.475	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
84	69.3.475	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
85	70.3.478	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
85	70.3.478	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
85	70.3.478	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
85	70.3.478	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
86	71.4.482	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
86	71.4.482	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
86	71.4.482	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
86	71.4.482	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
87	72.4.487	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
87	72.4.487	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
87	72.4.487	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
87	72.4.487	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
88	76.3.502	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ 1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
88	76.3.502	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ 1,\ 0,\ 2\big)$
89	77.3.507	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
89	77.3.507	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
89	77.3.507	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
89	77.3.507	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
89	77.3.507	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 3\big)$
89	77.3.507	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
89	77.3.507	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 3,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 3\big)$
89	77.3.507	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
90	78.3.512	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$
90	78.3.512	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -3,\ \sqrt{3},\ 3\big)$
90	78.3.512	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -1,\ -\sqrt{3},\ 1\big)$
90	78.3.512	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ -1,\ \sqrt{3},\ 1\big)$
90	78.3.512	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ \sqrt{3},\ 2,\ 0\big) + u_2\big(\sqrt{3},\ -1,\ 0,\ 2\big)$
90	78.3.512	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ -\sqrt{3},\ 1\big)$
90	78.3.512	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(1,\ -\sqrt{3},\ 2,\ 0\big) + u_2\big(-\sqrt{3},\ -1,\ 0,\ 2\big)$
90	78.3.512	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\ 1,\ 0,\ 0\big) + u_2\big(0,\ 0,\ \sqrt{3},\ 1\big)$
91	79.3.517	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ -3,\ -\sqrt{3},\ 3\big)$

No.	cSLG label	Néel direction	Complete allowed space
91	79.3.517	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
91	79.3.517	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
91	79.3.517	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
91	79.3.517	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-\sqrt{3},\, 3,\, 0,\, 0\big) + u_2\big(0,\, 0,\, -\sqrt{3},\, 3\big)$
91	79.3.517	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\, \sqrt{3},\, 2,\, 0\big) + u_2\big(\sqrt{3},\, 1,\, 0,\, 2\big)$
91	79.3.517	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(\sqrt{3},\, 3,\, 0,\, 0\big) + u_2\big(0,\, 0,\, \sqrt{3},\, 3\big)$
91	79.3.517	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u_1\big(-1,\, -\sqrt{3},\, 2,\, 0\big) + u_2\big(-\sqrt{3},\, 1,\, 0,\, 2\big)$
92	80.4.523	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
92	80.4.523	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
92	80.4.523	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
92	80.4.523	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$
92	80.4.523	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -3,\, -\sqrt{3},\, 3\big)$
92	80.4.523	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, -1,\, -\sqrt{3},\, 1\big)$
92	80.4.523	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -3,\, \sqrt{3},\, 3\big)$
92	80.4.523	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, -1,\, \sqrt{3},\, 1\big)$